

# Choroidal Vascularity and Axial Length Elongation in Highly Myopic Children: A 2-Year Longitudinal Investigation

Meng Xuan,<sup>1,2</sup> Decai Wang,<sup>1,2</sup> Ou Xiao,<sup>1,2</sup> Xinxing Guo,<sup>3</sup> Jian Zhang,<sup>1,2</sup> Qiuxia Yin,<sup>1,2</sup> Wei Wang,<sup>1,2</sup> Mingguang He,<sup>1,4,5</sup> and Zhixi Li<sup>1,2</sup>

<sup>1</sup>State Key Laboratory of Ophthalmology, Zhongshan Ophthalmic Center, Sun Yat-Sen University, Guangzhou, China

<sup>2</sup>Guangdong Provincial Key Laboratory of Ophthalmology and Visual Science, Guangdong Provincial Clinical Research Center for Ocular Diseases, Guangzhou, China

<sup>3</sup>Wilmer Eye Institute, Johns Hopkins University, Baltimore, Maryland, United States

<sup>4</sup>School of Optometry, The Hong Kong Polytechnic University, Kowloon, Hong Kong

<sup>5</sup>Research Centre for SHARP Vision, The Hong Kong Polytechnic University, Kowloon, Hong Kong

<sup>6</sup>Centre for Eye and Vision Research, Hong Kong, Hong Kong

Correspondence: Wei Wang, State Key Laboratory of Ophthalmology, Zhongshan Ophthalmic Center, Sun Yat-Sen University, No. 7 Jinsui Road, Guangzhou, China; [wangwei@gzoc.com](mailto:wangwei@gzoc.com).

Mingguang He, A034, School of Optometry, The Hong Kong Polytechnic University, Kowloon, Hong Kong; [mingguang.he@polyu.edu.hk](mailto:mingguang.he@polyu.edu.hk).

Zhixi Li, State Key Laboratory of Ophthalmology, Zhongshan Ophthalmic Center, Sun Yat-Sen University, No. 7 Jinsui Road, Guangzhou, China; [lsx11-11@163.com](mailto:lsx11-11@163.com).

MX and DW contributed equally to the work presented here and should therefore be regarded as equivalent authors.

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**PURPOSE.** To examine the influence of subfoveal choroidal thickness (SFCT) and choroidal vascularity index (CVI) on axial length (AL) elongation over a 2-year period in highly myopic children.

**METHODS.** In this prospective, longitudinal, observational study, 163 participants (74%), who were 8 to 18 years of age with bilateral high myopia (sphere  $\leq -6.0$  D) and without pathologic myopia, completed follow-up visits over 2 years. All participants underwent baseline and follow-up ocular examinations, including swept-source optical coherence tomography (SS-OCT) and AL measurements. SFCT and CVI were derived from SS-OCT scans using a deep-learning-based program for choroidal structure assessment.

**RESULTS.** The mean age of the participants at baseline was 15.0 years ( $\pm 2.3$ ), with males constituting 47% of the cohort. An inverse relationship was observed between AL elongation and increases in baseline age, baseline SFCT, and CVI, as well as a decrease in baseline AL. Adjusting for other factors, every 10- $\mu$ m increase in SFCT and each 1% increase in CVI were associated with decreases in AL elongation of 0.007 mm (95% confidence interval [CI],  $-0.013$  to  $-0.002$ ;  $P = 0.011$ ) and 0.010 mm (95% CI,  $-0.019$  to  $0.000$ ;  $P = 0.050$ ), respectively. The incorporation of SFCT or CVI into predictive models improved discrimination over models using only age, gender, and baseline AL (both  $P < 0.05$ , likelihood ratio test).

**CONCLUSIONS.** Our findings suggest a possible association between a thinner choroid and increased AL elongation over 2 years in children with high myopia, after adjusting for potential baseline risk factors such as age, gender, and initial AL.

**Keywords:** axial length, high myopia, choroidal thickness, choroidal vascularity index

The choroid, a highly vascularized tissue composed of vessels encased in stroma, is crucial for retinal and visual function by supplying nutrients to retinal pigment epithelial cells and the outer retina.<sup>1,2</sup> Swept-source optical coherence tomography (SS-OCT) enables noninvasive in vivo imaging of the choroidal structure at a micrometer resolution.<sup>3</sup> Utilizing choroidal thickness (CT) and the choroidal vascular index (CVI), derived from OCT images, has greatly contributed to the analysis of the choroidal structure, with CVI serving as a measurement of the vascular status of the choroid.<sup>4</sup>

Studies have consistently shown that highly myopic eyes tend to have a significantly thinner choroid and compromised choroidal circulation compared to normal eyes.<sup>5–16</sup> It is widely accepted that the choroid undergoes mechanical stretching and thinning due to axial elongation of the eye globe.<sup>17</sup> However, recent evidence from both human and animal research indicates that the choroid plays an important role in the processes of emmetropization and myopic progression.<sup>18–21</sup> SFCT exhibited a decrease in myopic children experiencing rapid refractive progression.<sup>18,20,22</sup> Moreover, normal chicks 12 to 14 days old, raised in temperature-

controlled brooders with a 12-hour light/dark cycle (8:30 AM to 8:30 PM), displayed faster eye growth with initially thinner choroids than those with thicker choroids.<sup>19</sup> However, these findings were limited to nonmyopic or low-to-moderate myopic children or animals, and whether thinner CTs could serve as predictive factors for the subsequent development of high myopia in children remained unknown. To bridge the existing knowledge gap, this study aimed to investigate whether a thinner CT or lower CVI at baseline could serve as predictors for subsequent 2-year axial length (AL) elongation.

## METHODS

### Participants

The study cohort was drawn from the Zhongshan Ophthalmic Centre, as detailed in prior publications.<sup>23</sup> Participants exhibiting bilateral high myopia (spherical equivalent  $\leq -6.00$  diopter [D]) were initially enrolled during the baseline investigation conducted from 2011 to 2013, with subsequent follow-up assessments conducted biennially. SS-OCT imaging procedures were performed during the initial follow-up visit spanning the years 2014 to 2015. The eligible participants for the present study were individuals 8 to 18 years of age who attended both the 2014 to 2015 and the 2016 to 2017 visits and demonstrated sufficient OCT image quality. Exclusion criteria included the presence of secondary causes of myopia, prior intraocular or refractive surgeries, or a history of severe eye disorders such as diabetic retinopathy, pathologic myopia, and uveitis.

The present study received ethical approval from the institutional review board of Zhongshan Ophthalmic Center, Sun Yat-Sen University, Guangzhou, China (identifier, 2012KYNL002). Written informed consent was obtained from all participants prior to their participation, and the study was conducted in compliance with the tenets of the Declaration of Helsinki.

### Ophthalmic Examinations

All participants underwent comprehensive ophthalmic examinations at baseline and all follow-up visits. AL measurements were obtained using optical low-coherence reflectometry (Lenstar LS 900; Haag-Streit AG, Köniz, Switzerland). In cases where the AL exceeded 32 mm, surpassing the valid measurement range of the Lenstar device, the IOLMaster (Carl Zeiss Meditec, Jena, Germany) was employed instead. Autorefraction was conducted utilizing an autorefractor device (KR8800; Topcon, Tokyo, Japan) following bilateral pupil dilation.

### Fundus Photography and Classification of Myopic Maculopathy

In this study, digital color fundus photographs were obtained using a CX-1 Retinal Camera (Canon, Tokyo, Japan). Each image featured a 45° field of view, centered on both the macula and the optic disc for each eye. For the purposes of this analysis, only images centered on the macula were utilized. Imaging was performed subsequent to the induction of complete mydriasis in each visit. Pupil dilation was achieved through the administration of compound tropicamide eye drops, each containing 5 mg/mL of tropicamide and 5 mg/mL of phenylephrine hydrochloride,

applied three times at 5-minute intervals. To ensure sufficient pupil dilation for fundus photography, a waiting period of 10 to 20 minutes was deemed necessary. Participants exhibiting pathological myopia (PM), as defined by the meta-PM classification criteria,<sup>24</sup> were excluded from the study.

### OCT Examination and Image Analysis

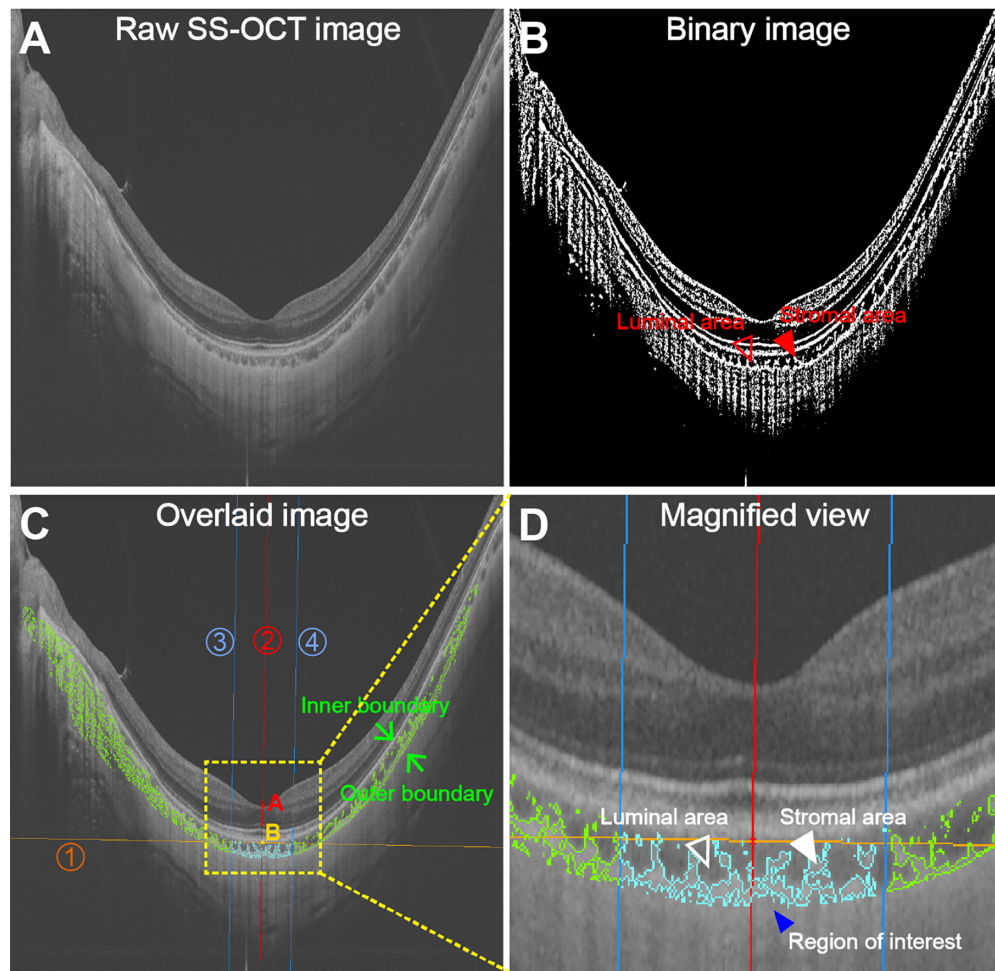
Choroidal imaging was performed using the SS-OCT (DRI OCT-1; Topcon), which employed a 1050-nm wavelength light source and a scanning speed of 100,000 A-scans per second. The scanning procedure and instrument parameters have been previously described in detail.<sup>25</sup> In brief, a 12-line radial scan pattern ( $12 \times 12$  mm) centered on the fovea was employed. The horizontal scans were utilized for the analysis of choroidal thickness and choroidal vascularity. All OCT scans were performed with the participants' pupils dilated and under standardized mesopic lighting conditions.

Participants with OCT scan quality scores  $>60$  were included in the final analysis. Choroidal parameters, including CT and CVI (the ratio of luminal area to total choroidal area) were obtained using a fully automated deep-learning choroidal structure analysis system developed and validated by our research group (Fig. 1).<sup>26</sup> The region of interest was defined as the macular region centered on the fovea with a diameter of 1500  $\mu\text{m}$ .<sup>4,27,28</sup> Details of this automated system have been described elsewhere in detail.<sup>26</sup> A validation study revealed a strong correlation between the automatic measurements and manual measurements.<sup>26</sup> Additionally, the automated measurements could reduce the intra-observer, interobserver, and intersession variations to a small degree.<sup>26</sup> All segmented choroidal boundaries underwent meticulous examination by an experienced clinician who performed manual segmentation as deemed necessary.

### Statistical Analysis

Participants with OCT scan quality scores  $>60$  were included in the final analysis. Due to the strong correlation observed between the AL measurements of the right and left eyes, statistical analysis was performed using data from the right eye only for each participant. To demonstrate AL elongation across different factors, the sample was divided into several subgroups based on baseline age, AL, CVI, and SFCT. For baseline age, participants were categorized into three groups: individuals with a baseline age of less than 13 years, participants with a baseline age between 13 and 15 years, and participants with a baseline age of 16 to 17 years. Regarding AL, the sample was categorized into three groups based on baseline AL measurements, following methodologies from prior research.<sup>29</sup> Group 1 included eyes with a baseline AL  $\leq 26.5$  mm, Group 2 included eyes with a baseline AL  $> 26.5$  mm but not exceeding 28.5 mm, and Group 3 included eyes with a baseline AL  $> 28.5$  mm. For SFCT and CVI, the classification was based on their corresponding 25% and 75% percentiles. By dividing the sample into these subgroups, we aimed to investigate the impact of various factors on AL elongation and observe any significant trends or differences among the groups.

Spherical equivalence refraction (SER) was equivalent to the sum of the spherical power and half of the cylindrical power. Continuous data were presented as means with standard deviations, and categorical data were reported as numbers with percentages. The participants' characteris-



**FIGURE 1.** Representative SS-OCT image analysis using an automated deep-learning choroidal analysis system. (A) Raw SS-OCT image. (B) Converted binary image obtained through the automated deep-learning choroidal analysis system, with luminal area and stromal area indicated. (C) Overlaid image with annotations. Point A represents the fovea center, and point B indicates the point on the choroidal inner boundary closest to the fovea center. Lines 1 to 4 represents the tangent of the inner boundary at point B (line 1), the perpendicular line (line 2) to line 1 at point B, and the boundaries on both sides (lines 3 and 4) of the 1500-μm-width region centered on the fovea, respectively. The green arrows indicate the inner and outer boundaries of the choroid. (D) A magnified view of the choroid in the foveal region. The 1500-μm-width choroidal region of interest is indicated by a closed blue arrowhead, the choroidal luminal area is indicated by a closed empty arrowhead, and the choroidal stromal area is labeled by a closed white color filled arrowhead.

tics were compared using unpaired *t*-tests for continuous parameters such as age, AL, SER, and choroidal parameters, and  $\chi^2$  tests for categorical parameters such as gender. For comparisons involving three groups, including baseline age, AL, SFCT, and CVI groups, one-way analysis of variance (ANOVA) was employed, followed by Tukey's post hoc test for further analysis. Univariate and multivariate regression models were utilized to determine factors associated with AL elongation. In the univariable analysis, factors with a significance level of  $P < 0.10$  were included in the multivariable analysis. Due to strong collinearity between AL and SER, only AL was selected and retained in the multivariable analysis.

The likelihood ratio test was employed to evaluate the adjusted  $R^2$  differences between two models, assessing whether the inclusion of SFCT or CVI enhanced the predictive value relative to a traditional model. This traditional model was based solely on established factors: age, gender, and AL. Estimated coefficients (mean differences), 95% confidence intervals (CIs), and two-sided *P* values were calcu-

lated. Statistical analyses were conducted using Stata 17.0 (Stata Corporation, College Station, TX, USA). A significance level of  $P < 0.05$  was used to determine statistical significance.

## RESULTS

### Participant Characteristics

Out of the initial 221 participants below 18 years of age who underwent SS-OCT examinations at baseline, 58 were excluded from the analysis. These exclusions were due to either missing the 2-year follow-up visit ( $n = 56$ ) or having suboptimal image quality ( $n = 2$ ). Finally, a total of 163 participants (73.76%) were included in the analysis (Fig. 2). In the SS-OCT analysis of 163 participants included in the final study, choroidal boundaries in two images (1.23%) required manual correction due to mis-segmentation by the automated program. The baseline characteristics of the included participants, including age, AL, SER, and choroidal



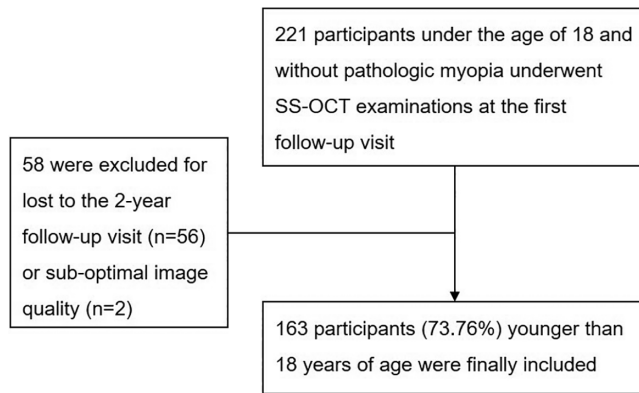


FIGURE 2. The workflow of participant selection for the study.

parameters, are summarized in Table 1. There were no significant differences observed between the included participants and those excluded from the analysis in terms of these baseline characteristics (Table 1).

### Distribution of AL Elongation

Table 2 presents the profiles of AL elongation over a 2-year period, categorized by age, gender, AL, SFCT, and CVI. In the 8- to 12-year-old age group, AL elongation over this 2-year period was observed to be greater than in the 13- to 15-year-old and 16- to 17-year-old age groups, with mean measurements of  $0.42 \pm 0.27$  mm,  $0.31 \pm 0.20$  mm, and  $0.19 \pm 0.18$  mm, respectively. Similarly, the group with AL measurements of 28.5 mm and above (Group 3) showed greater AL elongation compared to the groups with AL measurements greater than 26.5 mm and up to 28.5 mm (Group 2) and those with AL measurements of 26.5 mm or less (Group 1), with averaged measurements of  $0.37 \pm 0.23$  mm,  $0.28 \pm 0.21$  mm, and  $0.25 \pm 0.23$  mm, respectively. Additionally, eyes with thicker SFCT or higher CVI exhibited shorter AL elongation compared to eyes with thinner SFCT or lower CVI. No statistically significant gender differences were observed in AL elongation (Table 2).

### Risk Factors Associated With AL Elongation

A reduction in AL elongation was observed with increasing baseline age, decreasing baseline AL, and increases in baseline SFCT and CVI (Table 3). After adjusting for baseline age and gender, no significant association was found between baseline AL and AL elongation over a 2-year period.

TABLE 2. Axial Length Elongation Over 2 Years by Baseline Characteristics

| Baseline Characteristic   | AL Elongation (mm) |                 | P      |
|---------------------------|--------------------|-----------------|--------|
|                           | n                  | Mean $\pm$ SD   |        |
| Age (y)                   |                    |                 | <0.001 |
| 8–12                      | 28                 | $0.42 \pm 0.27$ |        |
| 13–15                     | 69                 | $0.31 \pm 0.20$ |        |
| 16–17                     | 66                 | $0.19 \pm 0.18$ |        |
| Gender                    |                    |                 | 0.283  |
| Female                    | 87                 | $0.26 \pm 0.22$ |        |
| Male                      | 76                 | $0.30 \pm 0.23$ |        |
| AL (mm)                   |                    |                 | 0.134  |
| Group 1 ( $\leq 26.5$ )   | 48                 | $0.25 \pm 0.23$ |        |
| Group 2 ( $> 26.5$ –28.5) | 97                 | $0.28 \pm 0.21$ |        |
| Group 3 ( $> 28.5$ )      | 18                 | $0.37 \pm 0.23$ |        |
| SFCT ( $\mu$ m)           |                    |                 | 0.052  |
| $\leq 170$                | 41                 | $0.35 \pm 0.25$ |        |
| $> 170$ –252              | 81                 | $0.26 \pm 0.20$ |        |
| $> 252$                   | 41                 | $0.24 \pm 0.22$ |        |
| CVI (%)                   |                    |                 | 0.286  |
| $\leq 62$                 | 40                 | $0.31 \pm 0.19$ |        |
| $> 62$ –66                | 83                 | $0.29 \pm 0.23$ |        |
| $> 66$                    | 40                 | $0.22 \pm 0.22$ |        |

P values were computed using unpaired *t*-tests for comparisons involving two variables and one-way ANOVA analysis for comparisons involving three or more variables.

However, when controlling for baseline age, gender, and AL, it was noted that, for each 10- $\mu$ m increase in SFCT, there was a decrease in AL elongation of 0.007 mm (95% CI,  $-0.013$  to  $-0.002$ ). Incorporating SFCT into Model 1 significantly improved the adjusted  $R^2$  value from 0.133 to 0.163, representing a 22.6% enhancement in the predictive value over the traditional model, as confirmed by the likelihood ratio test ( $P = 0.009$ ). Similarly, after adjusting for the same covariates, each 1% increase in CVI corresponded to a reduction in AL elongation of 0.010 mm (95% CI,  $-0.019$  to 0.000), although this finding was marginally significant ( $P = 0.050$ ). Additionally, the inclusion of CVI raised the adjusted  $R^2$  from 0.133 to 0.148, which translates to an 11.3% improvement in predictive value over the traditional model, as demonstrated by the likelihood ratio test ( $P = 0.049$ ).

### DISCUSSION

In this prospective cohort study, our findings suggest a possible association between a thinner choroid and increased AL elongation over 2 years in children with high myopia, after adjusting for potential baseline risk factors such as

TABLE 1. Baseline Characteristics of Participants and Non-Participants

| Baseline Characteristic             | Included Participants (n = 163) | Excluded Participants (n = 58) | P     |
|-------------------------------------|---------------------------------|--------------------------------|-------|
| Age (y), mean $\pm$ SD              | $15.0 \pm 2.3$                  | $15.0 \pm 2.2$                 | 0.884 |
| Gender, n (%)                       |                                 |                                | 0.659 |
| Male                                | 76 (47)                         | 29 (50)                        |       |
| Female                              | 87 (53)                         | 29 (50)                        |       |
| AL (mm), mean $\pm$ SD              | $27.14 \pm 1.13$                | $27.37 \pm 1.11$               | 0.201 |
| SER (D), mean $\pm$ SD              | $-9.69 \pm 1.93$                | $-10.10 \pm 2.50$              | 0.208 |
| Choroidal parameters, mean $\pm$ SD |                                 |                                |       |
| SFCT ( $\mu$ m)                     | $211.10 \pm 60.70$              | $212.84 \pm 59.54$             | 0.853 |
| CVI (%)                             | $64.31 \pm 3.52$                | $64.97 \pm 3.49$               | 0.228 |

P values were calculated based on unpaired *t*-tests for continuous data and  $\chi^2$  tests for categorical data.

TABLE 3. Effect of Baseline Parameters on AL Elongation Over 2 Years

| Baseline Parameter              | Univariate              |          |        | Model 1                 |          |        | Model 2                 |          |        | Model 3                 |          |        |
|---------------------------------|-------------------------|----------|--------|-------------------------|----------|--------|-------------------------|----------|--------|-------------------------|----------|--------|
|                                 |                         |          |        |                         |          |        |                         |          |        |                         |          |        |
|                                 | $\beta$                 | (95% CI) | P      | $\beta$                 | (95% CI) | P      | $\beta$                 | (95% CI) | P      | $\beta$                 | (95% CI) | P      |
| Age (y)                         |                         |          |        |                         |          |        |                         |          |        |                         |          |        |
| 8–12                            | 1 (Ref.)                |          | —      | 1 (Ref.)                |          | —      | 1 (Ref.)                |          | —      | 1 (Ref.)                |          | —      |
| 13–16                           | −0.112 (−0.203, −0.021) |          | 0.017  | −0.125 (−0.203, −0.021) |          | 0.008  | −0.115 (−0.206, −0.024) |          | 0.014  | −0.131 (−0.223, −0.039) |          | 0.005  |
| 16–18                           | −0.225 (−0.318, −0.133) |          | <0.001 | −0.228 (−0.203, −0.021) |          | <0.001 | −0.223 (−0.314, −0.133) |          | <0.001 | −0.226 (−0.318, −0.135) |          | <0.001 |
| Gender                          |                         |          |        |                         |          |        |                         |          |        |                         |          |        |
| AL (per 1-mm increase)          | −0.037 (−0.106, 0.031)  |          | 0.283  | −0.015 (−0.078, 0.056)  |          | 0.664  | −0.026 (−0.092, 0.040)  |          | 0.437  | −0.016 (−0.083, 0.050)  |          | 0.623  |
| Group 1 (≤26.5)                 | 1 (Ref.)                |          | —      | 1 (Ref.)                |          | —      | 1 (Ref.)                |          | —      | 1 (Ref.)                |          | —      |
| Group 2 (>26.5–28.5)            | 0.029 (−0.047, 0.106)   |          | 0.448  | 0.011 (−0.063, 0.084)   |          | 0.773  | −0.015 (−0.090, 0.060)  |          | 0.692  | 0.004 (−0.070, 0.077)   |          | 0.923  |
| Group 3 (>28.5)                 | 0.121 (0.002, 0.241)    |          | 0.047  | 0.106 (−0.014, 0.225)   |          | 0.082  | 0.073 (−0.047, 0.193)   |          | 0.233  | 0.104 (−0.014, 0.222)   |          | 0.084  |
| SFCT (per 10- $\mu$ m increase) | −0.007 (−0.013, −0.002) |          | 0.010  | —                       |          | —      | −0.007 (−0.012, −0.002) |          | 0.011  | —                       |          | —      |
| CVI (per 1% unit increase)      | −0.010 (−0.019, 0.000)  |          | 0.050  | —                       |          | —      | —                       |          | —      | −0.009 (−0.018, 0.000)  |          | 0.054  |

For multivariate model 1, the covariates were baseline age, gender, and baseline AL; the adjusted  $R^2$  of Model 1 was 0.133. For multivariate model 2, the covariates were baseline age, gender, baseline AL, and baseline CVI; the adjusted  $R^2$  of Model 2 was 0.163. For multivariate model 3, the covariates were baseline age, gender, baseline AL, and baseline CVI; the adjusted  $R^2$  of Model 3 was 0.148.

age, gender, and initial AL. Additionally, SFCT and CVI may contribute to understanding AL elongation in this population, although their predictive value may be limited due to the borderline nature of these associations. Further studies involving larger populations are needed to validate these findings.

To the best of our knowledge, this study presents the first longitudinal evidence identifying the predictive value of CT for AL elongation in highly myopic children. These findings are supported by a body of previous cross-sectional studies, which consistently demonstrate that highly myopic eyes exhibit a significantly thinner choroid compared to normal eyes.<sup>5–16</sup> However, due to the inherent limitations of cross-sectional studies, it remains challenging to establish a definitive temporal relationship between choroidal thinning and the onset or progression of myopia. Our longitudinal data suggest that a thinner CT at baseline is predictive of greater AL elongation in children with high myopia.

The observed decrease in choroidal blood flow may potentially contribute to retinal dysfunction and visual impairment among individuals with high myopia.<sup>10</sup> However, previous research has predominantly focused on evaluating choroidal thickness in highly myopic eyes, with limited investigation into the impact of choroidal structure on AL elongation. CT alone may not provide a comprehensive understanding of choroidal structural changes influenced by diseases, as the choroid is comprised of vessels embedded within the stroma.<sup>30</sup> A reduction in CT could arise from shrinkage in blood vessels, stromal tissue, or a combination of both factors.

To overcome this limitation, CVI emerges as a novel and valuable OCT-based parameter for assessing choroidal vascularity. CVI represents the ratio of luminal area to the total choroidal area, encompassing both luminal and stromal components.<sup>28</sup> As a result, CVI can precisely detect the specific choroidal component responsible for the changes in CT. This marker has demonstrated its reliability in analyzing choroidal structure in normal and diseased states and has become a subject of significant interest in ocular research.<sup>4,30</sup> In this study, we employed deep-learning techniques to automatically calculate the CVI in SS-OCT images. Our findings revealed that each 1% increase in CVI was associated with a reduction in AL elongation of 0.010 mm (95% CI, −0.019 to 0.000), although this result was marginally significant ( $P = 0.050$ ). A larger sample size is necessary to confirm these findings, but our data suggest that impaired choroidal circulation may contribute to AL elongation in children with high myopia.

This study possesses several strengths, which include the following: (1) utilization of SS-OCT technology for capturing choroidal images, offering superior image clarity compared to the previous SD-OCT generation; (2) implementation of a fully automated algorithm based on deep-learning techniques for precise measurement and analysis of choroidal structural parameters, reducing variability associated with interobserver, intra-observer, and repeated measurements, particularly for extremely thin choroids, thereby enhancing measurement repeatability and stability; and (3) comprehensive and reliable data collected from each participant, encompassing measurements of AL, refraction, and SS-OCT.

Nevertheless, this study has several limitations. First, the measurement of choroidal structural parameters was limited to within a 1500- $\mu$ m range beneath the fovea. Future research should explore the topographical distribution characteristics of choroidal structural parameters within the

macular region. Second, due to the resolution limitations of OCT images, the automated algorithm used in this study for measuring choroidal structural parameters can only analyze the medium to large choroidal vessels. In the future, it would be valuable to combine these parameters with indicators obtained from OCT angiography, such as choroidal microvasculature density, to provide a more comprehensive assessment of the structural characteristics of the choroid in highly myopic eyes. Third, the current study exclusively utilized horizontal OCT scans to investigate the impact of SFCT and CVI on AL elongation over a 2-year period in highly myopic children. Further research is warranted to explore the topographic distributions of SFCT and CVI in highly myopic children, along with their potential associations with AL elongation. Fourth, a larger sample size and a longer follow-up period are necessary to confirm these findings.

In conclusion, this study suggests a possible association between a thinner choroid and increased AL elongation over 2 years in children with high myopia, after adjusting for potential baseline risk factors such as age, gender, and initial AL. Furthermore, a lower CVI might be associated with greater AL elongation. Additionally, although SFCT and CVI can aid in understanding AL elongation, their predictive value is somewhat limited due to the marginal nature of these associations. Further studies with larger cohorts are necessary to confirm these findings. These findings underscore the significance of choroidal thickness and circulation in the process of AL elongation.

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